Lesson 1: Patterns, Sequences and the General Term

Describing a pattern term by term, sorting sequences as arithmetic, geometric or neither, and seeing a sequence as a function.

1. Sequences and Terms

A **sequence** is an ordered list of numbers. Each number is a **term**. We write t_1 for the first term, t_2 for the second term, and t_n for the term in position n (the **general term**). The position n is always a natural number: $1, 2, 3, \ldots$

A sequence can be described two ways: by a rule that gets the *next* term from the *previous* one ("start at 5 and add 4"), or by a formula for t_n that gives any term straight from its position ($t_n = 4n + 1$).

Example 1

A coffee shop stacks paper cups in towers. A tower of 1 cup is 11 cm tall, 2 cups is 13 cm, 3 cups is 15 cm, and 4 cups is 17 cm.

(a) Complete the table.

(b) Describe how to get each term from the previous term.

n (cups)	1	2	3	4	5	6
t_n (cm)	11	13	15	17		

(c) State t_1, t_4 and t_6. $t_1 =$ _____ $t_4 =$ _____ $t_6 =$ _____

2. Arithmetic, Geometric, or Neither

- In an **arithmetic sequence** each term is the previous term **plus** a constant (the **common difference**, d). Example: $5, 9, 13, 17, \ldots$ has $d = 4$.
- In a **geometric sequence** each term is the previous term **times** a constant (the **common ratio**, r). Example: $3, 6, 12, 24, \ldots$ has $r = 2$.
- Many sequences are **neither**: the square numbers $1, 4, 9, 16, \ldots$, the primes $2, 3, 5, 7, 11, \ldots$, and the **Fibonacci sequence** $1, 1, 2, 3, 5, 8, \ldots$ (each term is the sum of the two before it).

To test a sequence, subtract each term from the next (are the differences all equal?) and divide each term by the one before it (are the ratios all equal?).

Example 2

Classify each sequence as arithmetic, geometric or neither. Give d or r when there is one.

(a) $40, 33, 26, 19, \ldots$

(b) $2, -6, 18, -54, \ldots$

(c) $1, 3, 6, 10, 15, \ldots$

(d) $64, 16, 4, 1, \ldots$

(e) $\frac{1}{2}, \frac{5}{4}, 2, \frac{11}{4}, \ldots$

(f) $7, 7, 7, 7, \ldots$

3. Finite and Infinite Sequences

A **finite sequence** has a last term: $4, 7, 10, \ldots, 31$. An **infinite sequence** goes on forever, shown by three dots at the end with no last term: $4, 7, 10, \ldots$

4. Using a Formula for the General Term

When a formula for t_n is given, substitute the position n to get the term. To decide whether a number is a term, set the formula equal to it and solve for n: the number is a term only if n is a natural number.

Example 3

A sequence has general term $t_n = 5n - 3$.

(a) Write the first four terms.

(b) Determine t_{30}.

(c) Is 92 a term of the sequence? If so, which one?

(d) Is 100 a term of the sequence?

Example 4

The general term of a sequence is $t_n = 3(2)^{n-1}$. Write the first five terms and classify the sequence.

5. A Sequence Is a Function

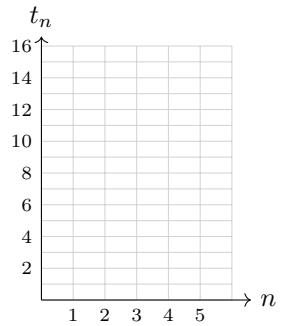
A sequence is a function whose **domain** is the natural numbers (the positions n) and whose **range** is the set of terms. Its graph is a set of separate points (n, t_n) — they are *not* joined, because there is no term at $n = 1.5$.

- An arithmetic sequence adds the same amount each step, so its points lie on a line: it is a **linear** function.
- A geometric sequence (with $r \neq 1$) multiplies each step, so its points curve: it is a **non-linear** (exponential) function.

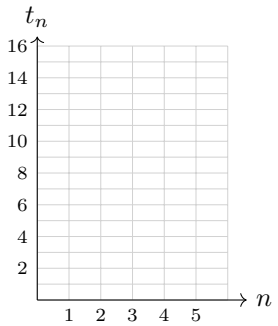
Example 5

Plot the first five terms of each sequence. State whether each is linear or non-linear.

(a) $2, 5, 8, 11, 14$



(b) $1, 2, 4, 8, 16$



Lesson 2: Arithmetic Sequences

Using the common difference and the general term $t_n = t_1 + (n-1)d$ to find a term, the number of terms, the first term or the common difference.

1. The Common Difference

In an **arithmetic sequence**, each term after the first is found by adding the same number to the term before it. That number is the **common difference**, d. To find d, subtract any term from the term that comes right after it:

$$d = t_2 - t_1 = t_3 - t_2 = t_4 - t_3 = \cdots = t_n - t_{n-1}$$

If $d > 0$ the terms increase; if $d < 0$ they decrease. Check every difference you can: one matching pair is not enough to call a sequence arithmetic. The terms may be fractions, decimals or algebraic expressions.

Example 1

Decide whether each sequence is arithmetic. If it is, state d and the next two terms.

(a) $13, 21, 29, 37, \ldots$

(b) $6, 1.5, -3, -7.5, \ldots$

(c) $\frac{1}{3}, \frac{5}{6}, \frac{4}{3}, \frac{11}{6}, \ldots$

(d) $3, 6, 12, 24, \ldots$

(e) $4a + b,\ 7a - b,\ 10a - 3b, \ldots$

(f) $2, 5, 9, 14, \ldots$

2. The General Term

Build the terms from t_1. Each step to the right adds one more d:

$$t_1, \quad t_2 = t_1 + d, \quad t_3 = t_1 + 2d, \quad t_4 = t_1 + 3d, \quad \ldots$$

The number of d's is always one less than the position, because it takes $n - 1$ steps to get from t_1 to t_n.

General term of an arithmetic sequence $\quad t_n = t_1 + (n-1)d$

t_n is the term in position n, t_1 is the first term (also written a), d is the common difference, n is the position (a natural number).

To write the general term of a given sequence, substitute t_1 and d, then expand and simplify.

Example 2

Consider the arithmetic sequence $41, 34, 27, 20, \ldots$

(a) Determine the general term in simplest form.

(b) Determine t_{30}.

(c) Which term is equal to -295?

(d) Is -300 a term of the sequence?

Example 3

(a) An arithmetic sequence has $t_4 = 11$ and $d = -3$. Write the first six terms.

(b) Fill in the missing terms of the arithmetic sequence with $d = 2.5$.

			4	

3. The Number of Terms

A finite sequence ends with a last term, and that last term is t_n. Substitute t_1, d and the last term into $t_n = t_1 + (n-1)d$, then solve for n. The answer must be a natural number.

To count the multiples of a number in an interval, find the smallest and the largest multiple in the interval. They are t_1 and t_n, and d is the number itself. "Inclusive" means the endpoints count; "between" means they do not.

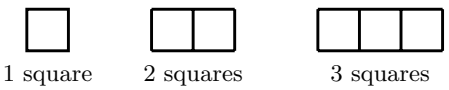
Example 4

(a) How many terms are in the sequence $8, 2, -4, \ldots, -166$?

(b) How many multiples of 9 are there between 100 and 600?

Example 5

Toothpicks are laid out to make a row of squares.



1 square 2 squares 3 squares

(a) Write the number of toothpicks for 1, 2, 3 and 4 squares, and the general term.

(b) How many toothpicks make a row of 40 squares?

(c) There are 200 toothpicks in a box. What is the longest row of squares that can be made, and how many toothpicks are left over?

4. Finding t_1 or d

The formula $t_n = t_1 + (n-1)d$ links four quantities: t_n, t_1, n and d. If you know any three of them, substitute and solve for the fourth.

Example 6

(a) An arithmetic sequence has $d = 6$ and $t_{15} = 71$. Determine t_1.

(b) An arithmetic sequence has $t_1 = 4$ and $t_{13} = -38$. Determine d and the general term.

(c) An arithmetic sequence has 16 terms. The first term is 12 and the last term is 87. Determine d.

5. Arithmetic Means

The terms between two non-consecutive terms of an arithmetic sequence are called **arithmetic means**. In $4, 10, 16, 22$, the arithmetic means between 4 and 22 are 10 and 16.

To insert k arithmetic means between two numbers, use the two numbers as t_1 and the last term. The sequence then has $k + 2$ terms. Find d, then build the means. One arithmetic mean between two numbers is simply their average.

Example 7

(a) Insert four arithmetic means between 7 and -18.

(b) Insert one arithmetic mean between -9 and 26.

6. Unknown Terms in an Arithmetic Sequence

If three expressions are consecutive terms of an arithmetic sequence, the two differences are equal:

$t_2 - t_1 = t_3 - t_2$ (equivalently, $t_2 = \dfrac{t_1 + t_3}{2}$: the middle term is the mean of its neighbours)

Set the differences equal, solve for the variable, then substitute to find the terms.

Example 8

The expressions $x - 4$, $2x + 1$ and $5x - 2$ are consecutive terms of an arithmetic sequence.

(a) Determine the value of x.

(b) State the three terms, the general term and t_{20}.

Lesson 3: Arithmetic Sequences in Context

Finding a sequence from two of its terms, modelling arithmetic growth and decay, and linking arithmetic sequences to linear functions.

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1. Two Terms That Are Not Consecutive

If two terms t_p and t_q are known (with $p < q$), you can still find d. Going from position p to position q takes $q - p$ steps, and each step adds d:

$d = \dfrac{t_q - t_p}{q - p}$ (change in the terms ÷ change in the positions)

Then substitute either known term into $t_n = t_1 + (n - 1)d$ to find t_1.

A second method: write both known terms with the general term formula. This gives a **system of two linear equations** in t_1 and d. Subtract one equation from the other to eliminate t_1.

Example 1

An arithmetic sequence has $t_4 = 19$ and $t_{11} = 54$.

(a) Determine d by counting steps.

(b) Determine t_1 and the general term.

(c) Solve the problem again using a system of equations.

Example 2

An arithmetic sequence has $t_6 = -8$ and $t_{20} = -43$. Determine d, t_1, the general term and t_{50}.

Example 3

Each row of an outdoor amphitheatre has the same number of extra seats as the row in front of it. Row 6 has 64 seats and row 16 has 94 seats.

(a) How many seats are in row 1?

(b) The back row has 133 seats. How many rows are there?

2. Arithmetic Growth

A quantity shows **arithmetic growth** when it increases by the same amount every period: a fixed raise each year, a fixed deposit each week, a set fee plus an hourly rate, and **simple interest**. With simple interest, interest is paid only on the original amount invested (the **principal**), so the interest is the same every year: interest per year = principal $\times$ annual rate.

Before writing a sequence, decide what t_1 stands for:

If t_1 is the value **at the start** (time 0), the value after k periods is t_{k+1}. If t_1 is the value **after the first period**, the value after k periods is t_k.

"When does it first pass ..." leads to an inequality. Solve it, round to the natural number that fits the question, and check the terms on both sides.

Example 4

Marco invests \$2400 at 3.75% simple interest per year.

(a) How much interest does he earn each year? Write the values at the end of years 1, 2 and 3.

(b) Let t_1 be the value at the end of year 1. Write the general term and find the value after 15 years.

(c) Redo (b) with $t_1 = 2400$, the value when he invests.

(d) After how many full years is the value first more than \$4000?

Example 5

A landscaping company charges a fixed equipment fee plus an hourly rate. A 1-hour job costs \$115, a 2-hour job costs \$160 and a 3-hour job costs \$205.

(a) State t_1 and d. What does d represent?

(b) Determine the fixed fee and the general term.

(c) What does an 8-hour job cost?

(d) A customer has \$600. What is the longest job, in whole hours, they can pay for?

3. Arithmetic Decay

A quantity shows **arithmetic decay** when it decreases by the same amount every period, so d is negative. Examples: equipment losing the same dollar value each year (**straight-line depreciation**), air temperature falling as altitude increases, and a candle burning down. The same care applies: state what t_1 means, and match the position to the time asked for.

Example 6

A food truck was bought new in 2018. It loses the same amount of value every year. In 2021 it was worth \$96 800, and in 2026 it was worth \$63 800. Let t_1 be its value in 2018.

(a) Determine the annual depreciation.

(b) Determine the purchase price.

(c) In what year will it first be worth less than \$15 000?

Example 7

A weather balloon is launched from a station where the air temperature is 21 °C. It records the temperature every 100 m as it rises. The temperature drops 0.65 °C for every 100 m. Let $t_1 = 21$, the reading at the ground.

(a) Write the first three readings and the general term.

(b) What is the temperature 2500 m above the station?

(c) At what height is the first reading below -10 °C?

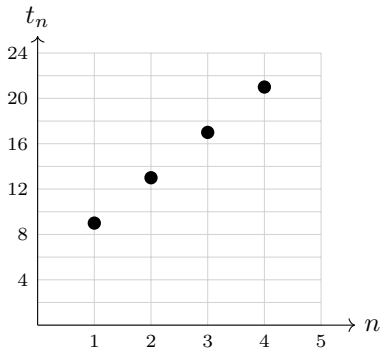
4. Arithmetic Sequences and Linear Functions

Plot the points (n, t_n) of an arithmetic sequence and they lie on a line: each step right by 1 goes up (or down) by d, so the **slope is** d. Expanding the general term shows the same thing:

$t_n = t_1 + (n-1)d = dn + (t_1 - d)$, which has the form $t_n = mn + b$ with $m = d$ and $b = t_1 - d$.

The value b is where the line crosses the vertical axis, one step before t_1. It is not a term: the domain is the natural numbers, so there is no point at $n = 0$ and the points are not joined. Going the other way, any linear function $y = mx + b$ with its domain restricted to the natural numbers gives an arithmetic sequence with $d = m$.

Example 8



The graph shows the first four terms of an arithmetic sequence.

(a) State t_1 and d, and write the general term in the form $t_n = mn + b$.

(b) What do the slope and the vertical intercept of the line through the points represent? Determine t_{25}.

(c) The function $f(x) = 30 - 2x$ has its domain restricted to the natural numbers. List the first four values of the range and show that they form an arithmetic sequence.

Lesson 4: Arithmetic Series

Adding the terms of an arithmetic sequence: pairing terms, the two sum formulas, working backwards from a sum, and recovering terms from a sum formula.

1. Sequences and Series

A **series** is the sum of the terms of a sequence. A sequence is written with commas; a series is written with plus signs. The arithmetic sequence $4, 9, 14, 19$ gives the **arithmetic series** $4 + 9 + 14 + 19$.

S_n is the sum of the **first** n **terms**: $S_1 = t_1$, $S_2 = t_1 + t_2$, $S_3 = t_1 + t_2 + t_3$, and so on. These running totals are called **partial sums**.

Example 1

Consider the arithmetic sequence $7, 12, 17, 22, 27, 32, \ldots$

(a) Write the series of the first six terms. _____

(b) Complete the table of partial sums.

(c) Calculate $S_4 - S_3$. Which term is it?

n	1	2	3	4	5	6
S_n						

2. Adding by Pairing

A long arithmetic series can be added quickly by writing it forwards and backwards and adding the two lines. Each column adds to the same total, $t_1 + t_n$, and there are n columns:

$$2S_n = n(t_1 + t_n).$$

This is the trick the young Carl Friedrich Gauss is said to have used to add $1 + 2 + 3 + \cdots + 100$ in moments.

Example 2

Use pairing to find the sum $5 + 10 + 15 + \cdots + 295 + 300$.

3. The Sum Formulas

Pairing works for every arithmetic series. Replacing t_n with $t_1 + (n-1)d$ gives a second version of the formula.

Sum of an arithmetic series $\quad S_n = \dfrac{n}{2}\left[2t_1 + (n-1)d\right] \quad$ or $\quad S_n = \dfrac{n}{2}(t_1 + t_n)$

S_n = sum of the first n terms, $\quad t_1$ = first term, $\quad d$ = common difference, $\quad n$ = number of terms, $\quad t_n$ = last term.

Use the first formula when d is known and the second when the last term is known. If the last term is given but n is not, find n first with $t_n = t_1 + (n-1)d$.

Example 3

Determine each sum.

(a) The first 25 terms of $7 + 11 + 15 + \ldots$

(b) The first 30 terms of $-20 + (-17.5) + (-15) + \ldots$

(c) S_{18} when $t_1 = -12$ and $t_{18} = 73$

Example 4

Determine the sum $52 + 45 + 38 + \cdots + (-39)$.

4. Working Backwards: Finding n, t_1 or d

Substitute everything you know into a sum formula and solve for the unknown.

- Finding t_1 or d gives a linear equation.
- Finding n gives a **quadratic equation**. Expand, move every term to one side, divide out any common factor, then factor. n must be a natural number, so reject a negative or fractional root. You can also make a table of values (with a calculator) of S_n for $n = 1, 2, 3, \ldots$ and look for the sum.

Example 5

How many terms of the series $6 + 10 + 14 + \ldots$ must be added to give a sum of 336?

Example 6

(a) An arithmetic series has $S_{20} = 650$ and $d = 3$. Determine t_1.

(b) An arithmetic series has 15 terms. The first term is 8 and the sum is 645. Determine the last term and d.

5. Word Problems

Read carefully to decide whether the question asks for one term (t_n: "how many in the last row") or a total (S_n: "how many altogether"). Define t_1, d and n in words before using a formula.

Example 7

Cans in a store display are stacked in rows. The bottom row has 30 cans and each row has 2 fewer cans than the row below it. The top row has 6 cans.

(a) How many rows are in the display?

(b) How many cans are in the display?

Example 8

A runner's weekly training distances form an arithmetic sequence. She runs 18 km in week 4 and 39 km in week 11.

(a) Determine d and t_1.

(b) Determine her total distance over the first 16 weeks.

6. Terms from a Sum Formula

Sometimes a formula for S_n is given instead of the terms. Each partial sum is the previous one plus one more term, so subtracting consecutive sums gives a term.

$t_1 = S_1$ and $t_n = S_n - S_{n-1}$ for $n \geq 2$

This also works when the sums S_n come from a table or a graph of points (n, S_n).

Example 9

An arithmetic series has $S_n = 3n^2 + 2n$.

(a) Determine the first four terms of the sequence.

(b) Determine t_{20}.

Lesson 5: Geometric Sequences

Multiplying by a common ratio: the general term $t_n = t_1 r^{n-1}$, finding any unknown from it, geometric means, and the curved graph.

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1. The Common Ratio

In a **geometric sequence** each term is the previous term multiplied by the same non-zero number, the **common ratio** r. Find r by dividing any term by the term before it:

Common ratio	$r = \dfrac{t_n}{t_{n-1}}$	$(n \geq 2)$	e.g. $r = \dfrac{t_2}{t_1} = \dfrac{t_3}{t_2}$

At least three terms are needed to decide the type of sequence: two terms such as $4, 8, \ldots$ fit an arithmetic sequence ($d = 4$) and a geometric sequence ($r = 2$). A negative r makes the signs alternate.

Example 1

State whether each sequence is arithmetic or geometric, give d or r, and write the next two terms.

(a) $5, 20, 80, \ldots$

(b) $405, 135, 45, \ldots$

(c) $-2, 6, -18, \ldots$

(d) $2, 2\sqrt{2}, 4, \ldots$

(e) $3a, 6a^2, 12a^3, \ldots$

(f) $50, 45, 40, \ldots$

2. The General Term

Starting from t_1, each step multiplies by r once more:

$$t_1, \quad t_2 = t_1 r, \quad t_3 = t_1 r^2, \quad t_4 = t_1 r^3, \quad \ldots$$

The exponent on r is always one less than the position.

General term of a geometric sequence $\quad t_n = t_1 r^{n-1}$
t_n = the term in position n, $\quad t_1$ = first term, $\quad r$ = common ratio, $\quad n$ = position (a natural number).

Leave the general term in the form $t_1 r^{n-1}$. Do not multiply t_1 by r first: $5(3)^{n-1}$ is *not* 15^{n-1}, because the exponent applies only to the 3. When t_1 is a power of r, the exponent laws combine them into a single power.

Example 2

Write the general term of each sequence, then determine the indicated term.

(a) $7, 14, 28, \ldots$; t_{12}

(b) $1250, -250, 50, \ldots$; t_8

(c) $\frac{2}{9}, \frac{2}{3}, 2, \ldots$; t_9

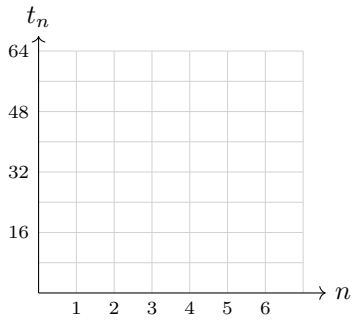
(d) $4, 8, 16, \ldots$; write t_n as a single power of 2.

A geometric sequence is a function of n with an exponent: its graph is **non-linear**. When $r > 1$ the points curve up steeply; when $0 < r < 1$ they curve down towards zero; when $r < 0$ they jump from one side of the n-axis to the other.

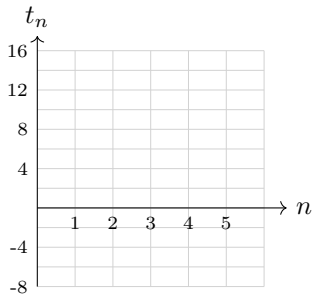
Example 3

Plot the first terms of each sequence and describe the graph.

(a) $t_n = 64(0.5)^{n-1}$, $n = 1$ to 6



(b) $t_n = (-2)^{n-1}$, $n = 1$ to 5



3. Finding the Number of Terms

To find the number of terms, set t_n equal to the last term and solve for n. Divide both sides by t_1, then write both sides as powers of the same base and compare exponents. (A table of values of $t_1 r^{n-1}$ for $n = 1, 2, 3, \ldots$ also works.)

Example 4

Determine the number of terms in each sequence.

(a) $3, 6, 12, \ldots, 6144$

(b) $1215, -405, 135, \ldots, -\frac{5}{9}$

4. Two Terms Known: Finding t_1 and r

If two terms are known, write each with the general term and **divide** the equations: t_1 cancels and a power of r is left. Then substitute r back to find t_1.

- An **odd** power has one real root: $r^3 = -8 \Rightarrow r = -2$.
- An **even** power has two: $r^2 = 9 \Rightarrow r = \pm 3$. Both values give a valid sequence, so there are two answers.

Example 5

In a geometric sequence, $t_3 = 12$ and $t_6 = -96$. Determine r, t_1 and the general term.

Example 6

In a geometric sequence, $t_2 = 12$ and $t_4 = 108$. Explain why there are two possible sequences, and determine t_1 and t_5 for each.

5. Geometric Means

The terms between two non-consecutive terms of a geometric sequence are called **geometric means**. To insert k means between two numbers, treat the numbers as t_1 and the last term of a sequence of $k + 2$ terms, find r, then multiply.

Example 7

(a) Insert three geometric means between 5 and 405.

(b) Insert two geometric means between 4 and -500.

6. Expressions in a Geometric Sequence

If a, b, c are consecutive terms of a geometric sequence, the ratios are equal:

$$\frac{b}{a} = \frac{c}{b} \qquad \text{which cross-multiplies to} \qquad b^2 = ac$$

Set up this equation with the expressions, solve, and check that no term is zero.

Example 8

Each set of expressions gives three consecutive terms of a geometric sequence. Determine x, the terms, and r.

(a) $x+1,\ x+4,\ x+10$

(b) $x,\ x+4,\ 9x$

Lesson 6: Geometric Growth and Decay

Modelling repeated percent increases and decreases with geometric sequences, deciding where the sequence starts, and finding the number of periods with a table of values.

1. Growth and Decay Factors

When a quantity changes by the **same percent** every period (every year, hour, bounce, ...), each value is the previous value times the same number. The values form a geometric sequence, and the common ratio r is called the **growth factor**. When $0 < r < 1$ the quantity shrinks, and we call it **geometric decay** (r is then also called the decay factor).

Increase by p% each period:	$r = 1 + \frac{p}{100}$	(up 6%: $r = 1.06$)
Decrease by p% each period:	$r = 1 - \frac{p}{100}$	(down 6%: $r = 0.94$)
Doubles: $r = 2$ Triples: $r = 3$ Halves: $r = \frac{1}{2}$	Keeps a fraction f of its value ("rebounds to $\frac{3}{5}$ of its height"): $r = f$	

Read carefully: "loses $\frac{1}{4}$ of its value" means it *keeps* $\frac{3}{4}$, so $r = \frac{3}{4}$; "rebounds to 80%" means $r = 0.8$, but "loses 80% of its height" means $r = 0.2$.

Example 1

State the growth factor r in each situation.

(a) A herd of elk grows by 7% each year.

(b) A laptop loses 28% of its value each year.

(c) The views of a video triple every day.

(d) A swing's arc gets shorter by one-eighth with each swing.

(e) A ball rebounds to 65% of its previous height.

(f) Wages rise by 2.75% per year.

2. Building a Growth or Decay Sequence

Multiply by r to get each next value. Keep the unrounded values in your calculator and round only the answers you write down. Once you know t_1 and r, any term comes from $t_n = t_1 r^{n-1}$.

Example 2

A community garden program had 450 members when it started in January 2022. Its membership grows by 12% each year.

(a) Complete the table (nearest whole number).

January	2022	2023	2024	2025
Members	450			

(b) Show that the sequence is geometric and state r.

(c) If t_1 is the membership in January 2022, which term gives the membership in January 2032?

(d) Estimate the membership in January 2032.

3. Where Does the Sequence Start?

Problems often ask for the value "after n years" or "after the nth bounce". There are two correct ways to set up the sequence; the danger is mixing them up. Let P be the starting amount (at time 0).

- **Start at time 0:** $t_1 = P$. Then t_2 is the value after 1 period, so the value after n periods is t_{n+1}.
- **Start after one period:** $t_1 = Pr$. Then the value after n periods is t_n.

Both give the same result:

Value after n periods $= Pr^n$ (P = starting amount at time 0, r = growth factor)

Example 3

A ball is dropped from a height of 16 m. Each bounce reaches 75% of the height of the bounce before. Determine the height of the 7th bounce, to the nearest centimetre, in two ways.

(a) Let $t_1 = 16$, the height it is dropped from. Which term is the height of the 7th bounce? Evaluate it.

(b) Let t_1 be the height of the first bounce. State t_1, then evaluate the term that gives the 7th bounce.

(c) Write a formula for the height h of the ball after x bounces. _____

4. Doubling, Half-Life and Other Periods

The change does not have to happen once a year. If a quantity doubles every 20 minutes, the period is 20 minutes, and the number of periods is the total time divided by 20 minutes. The **half-life** of a radioactive substance is the time it takes for half of it to decay, so $r = \frac{1}{2}$ for each half-life.

Number of periods $n = \dfrac{\text{total time}}{\text{length of one period}}$, amount $= Pr^n$

Example 4

(a) A culture starts with 300 bacteria and the number doubles every 20 minutes. How many bacteria are there after 4 hours?

(b) A hospital receives 96 mg of a medical isotope whose half-life is 6 hours. How much is left after 2 days?

5. Compound Interest

When interest is **compounded annually** at a rate i (as a decimal), the interest each year is added to the balance, and the next year's interest is paid on that new balance. Each year the value is multiplied by $1 + i$, so the values form a geometric sequence with $r = 1 + i$.

Compound interest: $A = P(1+i)^n$ P = principal (amount invested), i = interest rate per period (decimal), n = number of periods, A = final amount.

If interest is compounded more often, use the rate *per period* and count periods: 6% per year compounded monthly gives $i = \frac{0.06}{12} = 0.005$, and 3 years is $n = 36$ months.

Example 5

Lena invests \$6500 in a GIC that pays 3.8% per year, compounded annually.

(a) State the growth factor.

(b) Write a product (do not evaluate) for the value after 1 year and after 2 years.

(c) Let t_1 be the value after 1 year. Write t_n and show it equals $6500(1.038)^n$.

(d) Determine the value after 9 years, to the nearest cent.

Example 6

An investment of \$2500 grows to \$3400 in 7 years with interest compounded annually. Determine the annual interest rate, to the nearest tenth of a percent.

6. Finding the Number of Periods with a Table

To find *when* a growing or decaying quantity passes a target, we need the exponent n. Logarithms are not part of this course, so use a **table of values**: evaluate Pr^n on your calculator for $n = 1, 2, 3, \ldots$ (a graphing calculator's table does this for you). Find the first whole number of periods that passes the target, and check the value just before it.

Example 7

A quantity of water holds 500 mg of a contaminant. Each pass through a filter removes 30% of the contaminant that is left. How many passes are needed to bring it below 40 mg?

Example 8

A tractor bought for \$180 000 loses 22% of its value in the first year and 9% of its value each year after that.

(a) Determine its value after 1 year.

(b) Determine its value after 6 years, to the nearest dollar.

(c) After how many years is it first worth less than \$80 000?

Lesson 7: Geometric Series

Adding the terms of a geometric sequence with a formula, and using it to find a sum, the number of terms, the first term or the common ratio.

1. The Sum of a Geometric Series

A **geometric series** is the sum of the terms of a geometric sequence: $2, 6, 18, 54$ is a sequence; $2 + 6 + 18 + 54$ is a series. S_n is the sum of the first n terms.

To find a formula, write S_n, multiply every term by r, and subtract. Almost every term cancels:

$$\begin{array}{rcccccccccc} S_n &=& t_1 &+& t_1 r &+& t_1 r^2 &+& \cdots &+& t_1 r^{n-1} \\ rS_n &=& & & t_1 r &+& t_1 r^2 &+& \cdots &+& t_1 r^{n-1} + t_1 r^n \\ \hline rS_n - S_n &=& -t_1 & & & & & & & + & t_1 r^n \end{array}$$

So $S_n(r-1) = t_1(r^n - 1)$. Dividing by $r - 1$ gives the formula. Since $t_1 r^n = r(t_1 r^{n-1}) = rt_n$, there is a second form that uses the last term instead of n.

Sum of a geometric series ($r \neq 1$): $S_n = \dfrac{t_1(r^n - 1)}{r - 1}$ or $S_n = \dfrac{rt_n - t_1}{r - 1}$

Use the first form when you know n; use the second when you know the last term t_n. If $r = 1$, every term equals t_1, so $S_n = nt_1$.

Example 1

For the series $5 + 10 + 20 + 40 + 80$, find the sum by adding, by the first formula, and by the second formula.

2. Finding the Sum

Find t_1 and r (divide any term by the one before it), then decide which form fits. When r is negative, put it in brackets: $(-3)^8$, not -3^8. Give exact answers (fractions) unless a rounding is asked for.

Example 2

Determine each sum.

(a) The first 12 terms of $7+21+63+\cdots$

(b) The first 8 terms of $2-6+18-\cdots$

(c) The first 7 terms of $128+32+8+\cdots$ (exact)

Example 3

Determine each sum.

(a) $3 + 12 + 48 + \cdots + 12\,288$

(b) $1458 - 486 + 162 - \cdots + 2$

3. Finding n, t_1 or r

Substitute what you know into a formula and solve for the unknown.

- **Finding n:** isolate the power ($3^n = 729$) and find the exponent by trial, with a table of values. Or find the last term with the second formula, then its position with $t_n = t_1 r^{n-1}$.
- **Finding t_1:** use a known term and r ($t_1 = \frac{t_k}{r^{k-1}}$), or solve the sum formula for t_1.
- **Finding r:** when $n = 3$, $S_3 = t_1 + t_1 r + t_1 r^2$ gives a quadratic equation to factor (there may be two values of r). For larger n, try values of r in a table.

Example 4

(a) How many terms of $4 + 20 + 100 + \cdots$ add to 15 624?

(b) The series $4 - 8 + 16 - \cdots$ has a sum of 684. Determine the last term and the number of terms.

Example 5

(a) In a geometric sequence, $t_4 = 540$ and $r = 3$. Determine S_8.

(b) A geometric series has $t_1 = 6$ and $S_3 = 126$. Determine all possible values of r.

(c) A geometric series has $t_1 = 4$ and $S_6 = 1456$. Use a table to find r ($r > 0$), then find t_3.

4. Terms from a Sum Formula

S_n is the sum of the first n terms, and S_{n-1} is the sum of the first $n - 1$ terms. Their difference is the nth term.

$t_n = S_n - S_{n-1}$ (and $t_1 = S_1$)

Example 6

The sum of the first n terms of a series is $S_n = 2(5^n - 1)$.

(a) Determine the first four terms.

(b) Determine t_7 without finding the general term.

5. Geometric Series in Context

A dropped ball travels *down* and *up*. When it hits the ground for the kth time, it has fallen k times (the drop plus $k - 1$ falls after bounces) and risen $k - 1$ times. Find the two sums separately and add them.

Total distance at the kth hit = (sum of k falls, starting with the drop height) + (sum of $k - 1$ rises, starting with the first bounce height)

Example 7

A ball is dropped from a height of 27 m. Each bounce reaches $\frac{2}{3}$ of the height before.

(a) How high is the 5th bounce, to the nearest centimetre?

(b) Determine the total vertical distance the ball has travelled when it hits the ground for the 6th time, to the nearest centimetre.

Example 8

A club uses a phone tree for news. The president calls 5 members (level 1). Each of them calls 5 new members (level 2), and so on.

(a) How many members are called in the first 4 levels?

(b) How many levels are needed to reach at least 12 000 members?

Lesson 8: Infinite Geometric Series

Watching partial sums settle on a value, deciding when an infinite geometric series has a sum, and using $S_\infty = \dfrac{t_1}{1-r}$.

1. Partial Sums: Convergent or Divergent

An **infinite series** has no last term: $6 + 2 + \frac{2}{3} + \ldots$ We cannot add infinitely many terms one at a time, but we can add the first n terms. These sums $S_1, S_2, S_3, \ldots$ are the **partial sums** of the series.

- If the partial sums get closer and closer to one fixed number, the series is **convergent**, and that number is its **sum to infinity**, written S_∞.
- If the partial sums do not settle on one number (they grow without limit, or keep jumping), the series is **divergent** and has no sum.

Example 1

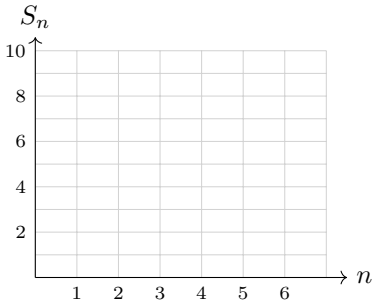
Two infinite geometric series are given. Series A: $6 + 2 + \frac{2}{3} + \frac{2}{9} + \ldots$ Series B: $5 + 10 + 20 + 40 + \ldots$

(a) State r for each series, then complete the table of partial sums (to 4 decimal places where needed).

Series A: $r =$ _____ Series B: $r =$ _____

n	1	2	3	4	5	6
S_n for Series A						
S_n for Series B						

(b) Plot the partial sums of Series A. What value do they approach?



(c) Classify each series as convergent or divergent. Give S_∞ where it exists.

2. The Sum to Infinity of a Geometric Series

Whether a geometric series converges depends on what happens to r^n as n gets large:

r	r^5	r^{20}	r^{50}	as n grows, r^n …
0.5	0.0313	0.0000…	0.0000…	gets closer and closer to 0
−0.5	−0.0313	0.0000…	0.0000…	gets closer and closer to 0
1.5	7.5938	3325.26	about 638 million	grows without limit
−1.5	−7.5938	3325.26	about 638 million	grows in size, sign keeps switching

Write the Lesson 7 formula as $S_n = \dfrac{t_1(1-r^n)}{1-r}$ (multiply its top and bottom by -1). When $-1 < r < 1$, r^n shrinks to 0, so S_n gets closer and closer to $\dfrac{t_1(1-0)}{1-r}$.

Infinite geometric series. If $-1 < r < 1$, the series is **convergent** and

$$S_\infty = \frac{t_1}{1-r}.$$

If $r \le -1$ or $r \ge 1$, the series is **divergent** and has no sum to infinity. (For $r = 1$ the partial sums $t_1, 2t_1, 3t_1, \ldots$ grow; for $r = -1$ they jump between t_1 and 0.)

Example 2

Determine r and decide whether each series is convergent or divergent. Calculate S_∞ where it exists.

(a) $45 + 15 + 5 + \ldots$

(b) $20 - 16 + 12.8 - \ldots$

(c) $2 + 5 + 12.5 + \ldots$

(d) $4 - 4 + 4 - 4 + \ldots$

(e) $\frac{3}{4} + \frac{1}{2} + \frac{1}{3} + \ldots$

(f) $0.006 + 0.06 + 0.6 + \ldots$

Example 3

Consider the series $40 + 30 + 22.5 + \ldots$

(a) Determine S_∞.

(b) Determine S_{10} to the nearest hundredth.

(c) Using a table of values, determine the smallest number of terms whose sum is within 1 of S_∞.

3. Finding t_1 or r

Substitute what you know into $S_\infty = \dfrac{t_1}{1 - r}$ and solve. Any value of r you find must satisfy $-1 < r < 1$, or the series has no sum.

Example 4

(a) $S_\infty = 50$ and $r = 0.3$. Determine t_1.

(b) $t_1 = 12$ and $S_\infty = 8$. Determine r.

(c) In an infinite geometric series, $t_1 = 64$ and $t_3 = 16$. Determine S_∞ for each possible series.

(d) Is there an infinite geometric series with $t_1 = 10$ and $S_\infty = 4$? Explain.

4. Repeating Decimals

A repeating decimal is an infinite geometric series. Split it into its repeating blocks: each block is the one before it moved one block-length to the right, so $r = 0.1$ for a 1-digit block, $r = 0.01$ for a 2-digit block, and so on. Then S_∞ gives the decimal as a fraction. If some digits do not repeat, keep them separate and add them at the end.

Example 5

Write each repeating decimal as an infinite geometric series, then as a fraction in lowest terms.

(a) $0.\overline{4}$ (b) $0.\overline{63}$ (c) $1.2\overline{7}$

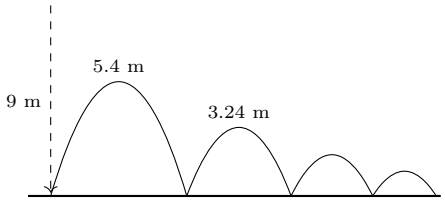
5. Applications

When a quantity keeps shrinking by the same percent (a bounce height, weekly sales, yearly output), the total over all time is S_∞. In a real situation the process eventually stops, so S_∞ is the largest the total could ever be.

For a **bouncing ball**, the ball travels *down* and *up*. Add two series: the drops (starting with the original height) and the rises (starting with the first rebound height).

Example 6

A ball is dropped from a height of 9 m. Each time it bounces, it rises to 60% of the height it fell from. Assume it keeps bouncing forever.



(a) Write the series for the distances travelled down, and for the distances travelled up.

(b) Determine the total vertical distance the ball travels.

Example 7

A new phone app is downloaded 12 000 times in its first week. Each week after that, the number of downloads is 85% of the week before.

(a) How many downloads are there in week 5?

(b) How many downloads are there in the first 10 weeks, to the nearest whole number?

(c) If this pattern went on forever, what is the greatest total number of downloads the app could reach?

(d) Why might the real total differ from (c)?

6. Series with a Variable Ratio

When the terms contain a variable, find r as an expression. The series converges only for the values of the variable that make $-1 < r < 1$: solve this double inequality by doing the same thing to all three parts.

Example 8

Consider the infinite geometric series $3 + \dfrac{3x}{2} + \dfrac{3x^2}{4} + \ldots$

(a) Determine r and the values of x for which the series converges.

(b) Write an expression for S_∞ and evaluate it when $x = 1$.

(c) Determine the value of x for which $S_\infty = 4$.

Lesson 9: Unit 1 Review

Every formula of the unit on one page, how to choose between them, and mixed practice in the style of the unit test.

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1. Unit Summary

	Arithmetic	**Geometric**
Pattern	add the same number: $d = t_2 - t_1$	multiply by the same number: $r = \dfrac{t_2}{t_1}$
General term	$t_n = t_1 + (n-1)d$	$t_n = t_1 r^{n-1}$
Graph of (n, t_n)	points on a line; slope $= d$ (linear)	points on a curve (non-linear, exponential)
Sum of n **terms**	$S_n = \dfrac{n}{2}\left[2t_1 + (n-1)d\right] = \dfrac{n}{2}(t_1 + t_n)$	$S_n = \dfrac{t_1(r^n - 1)}{r - 1} = \dfrac{rt_n - t_1}{r - 1},\ r \neq 1$
Sum to infinity	does not exist (unless every term is 0)	$S_\infty = \dfrac{t_1}{1 - r}$ only if $-1 < r < 1$ (convergent); otherwise divergent
Three terms a, b, c **in a row**	$b - a = c - b$	$\dfrac{b}{a} = \dfrac{c}{b}$, so $b^2 = ac$
k **means between two terms**	$k + 1$ steps of d	$k + 1$ steps of r (even number of steps: r can be $\pm$)
Growth / decay	a fixed amount each time (simple interest, fixed drop)	a fixed percent each time: $r = 1 +$ rate or $r = 1 -$ rate

How to choose

1. **Add or multiply?** A fixed amount each step is arithmetic. A fixed percent or a fixed multiple (doubling, halving, "rises to 70%") is geometric.
2. **One value or a total?** One value ("in year 8", "row 20", "height of the 5th bounce") uses t_n. A total ("altogether", "in the first 10 weeks", "total distance") uses S_n. A total that goes on forever uses S_∞.
3. **Count the terms carefully.** If t_1 is the starting value (time 0), the value after n years is $t_{n+1} = t_1 r^n$. If t_1 is the value after year 1, the value after n years is t_n. To count terms from first to last: $n = \dfrac{t_n - t_1}{d} + 1$.
4. **Finding** n**.** From t_n (arithmetic): a linear equation. From S_n (arithmetic): a quadratic — factor it, or use a table of values. Geometric: use a table of values (trial).
5. **Given a formula for** S_n**:** $t_1 = S_1$ and $t_n = S_n - S_{n-1}$.
6. **Bouncing ball:** add the series of drops and the series of rises (the rises start with the first rebound).

2. Choosing the Formula

Example 1

For each situation, decide whether it is arithmetic or geometric, and whether it asks for a term or a sum. Then answer it.

(a) A theatre has 24 seats in the first row, and each row has 3 more seats than the row in front of it. How many seats are in the first 20 rows?

(b) A car bought for \$32 000 loses 15% of its value each year. What is it worth 6 years after it was bought?

(c) A message is sent to 3 people in round 1. In each round after that, everyone who just got it sends it to 3 new people. How many people get the message in the first 8 rounds?

(d) A pendulum's first swing is 50 cm long and each swing after that is 90% as long as the one before. What total distance does it travel if it swings forever?

3. Working with Terms

Example 2

In an arithmetic sequence, $t_5 = 17$ and $t_{12} = 45$.

(a) Determine d, t_1 and the general term.

(b) Determine S_{30}. Is 205 a term of the sequence?

Example 3

In a geometric sequence, $t_2 = -12$ and $t_5 = 324$. Determine r, t_1, the general term and S_8.

Example 4

(a) Determine k so that $k+4$, $3k$ and $4k+2$ are consecutive terms of an arithmetic sequence. State the terms.

(b) Determine x so that $x-2$, $x+2$ and $x+10$ are consecutive terms of a geometric sequence. State the terms and r.

Example 5

The sum of the first n terms of a sequence is $S_n = 2n^2 + 3n$. Determine t_1, t_2 and t_{10}, and identify the sequence.

4. Finding the Number of Terms

Example 6

(a) How many terms of $3 + 5 + 7 + \ldots$ add to 255?

(b) How many terms of $5 + 10 + 20 + \ldots$ add to 5115?

Example 7

A town has a population of 8400 and is growing by 3.5% per year.

(a) Predict the population 5 years from now.

(b) After how many full years will the population first be over 12 000?

5. Series in Context

Example 8

A ball is dropped from a height of 15 m. After each bounce it rises to 70% of the height it fell from.

(a) How high, to the nearest hundredth of a metre, does it rise after the 4th bounce?

(b) What total vertical distance has it travelled when it hits the ground for the 5th time?

(c) What total vertical distance does it travel if it bounces forever?